

'DIVYA' SYSTEM

*Fault Detection and Isolation System for India's
Low Voltage Distribution Grid*



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INTRODUCTION:

India's electrical safety is at risk. Every year, India has about 24,000 electrical mishaps, resulting in approximately 1,000 electrocution deaths, or about three deaths per day, with figures growing by more than 50% in the last decade. [International Journal of Electrical Power and Energy Systems]. A large percentage of these occurrences are caused by wire breaks, inadequate infrastructure, and delayed fault identification. This pressing issue emphasizes the need for a low-cost, real-time, automated wire break detection and isolation system to make electricity distribution safer, more dependable, and more responsive to fault circumstances in both urban and rural settings.

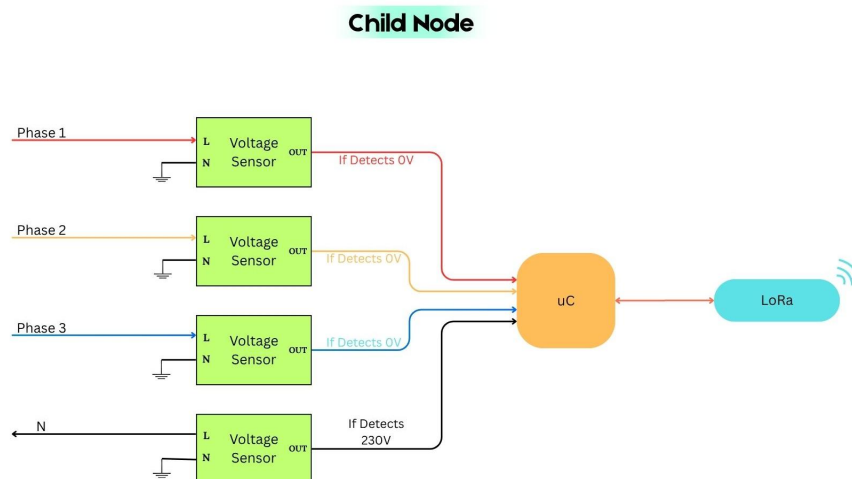
We propose a technical solution to this problem. The 'Divya System' is a novel system for detecting wire breakages and defects in three-phase electrical distribution lines via IoT and wireless communication. We will utilize a simple two-tier system. One will detect the malfunction, while another will switch off the power. They will be connected using LoRa, a wireless, fast, and low-cost communication technology.

The approach is simple yet efficient, fast and inexpensive, making it the most dependable alternative for usage in large-scale systems across India. This method has the potential to make India a safer place to live. Our system will protect the public from the dangers of overhead low-voltage electrical distribution lines.

DESIGN LOGIC & RATIONALE:

Unpredictable weather sometimes becomes a nightmare to the public. Not only does it have a disaster effect but also it can tear low voltage distribution lines making it a silent killer no one even gets a warning before. Guess what? That's a very common situation throughout India. We need a system which will cut-off the supply at the time the disruption happens. Divya System is two tier safety system one for detection and another for protection. They are Child Node and Parent Node.

1. **Child Node:** When disruptions happen there must be a voltage spike or huge imbalance in the supply line. We used a continuous voltage measuring circuit which continuously measures voltage in the line. A micro-controller is used to monitor the voltage. At the time it finds a disruption it will send a distress call to its parent.



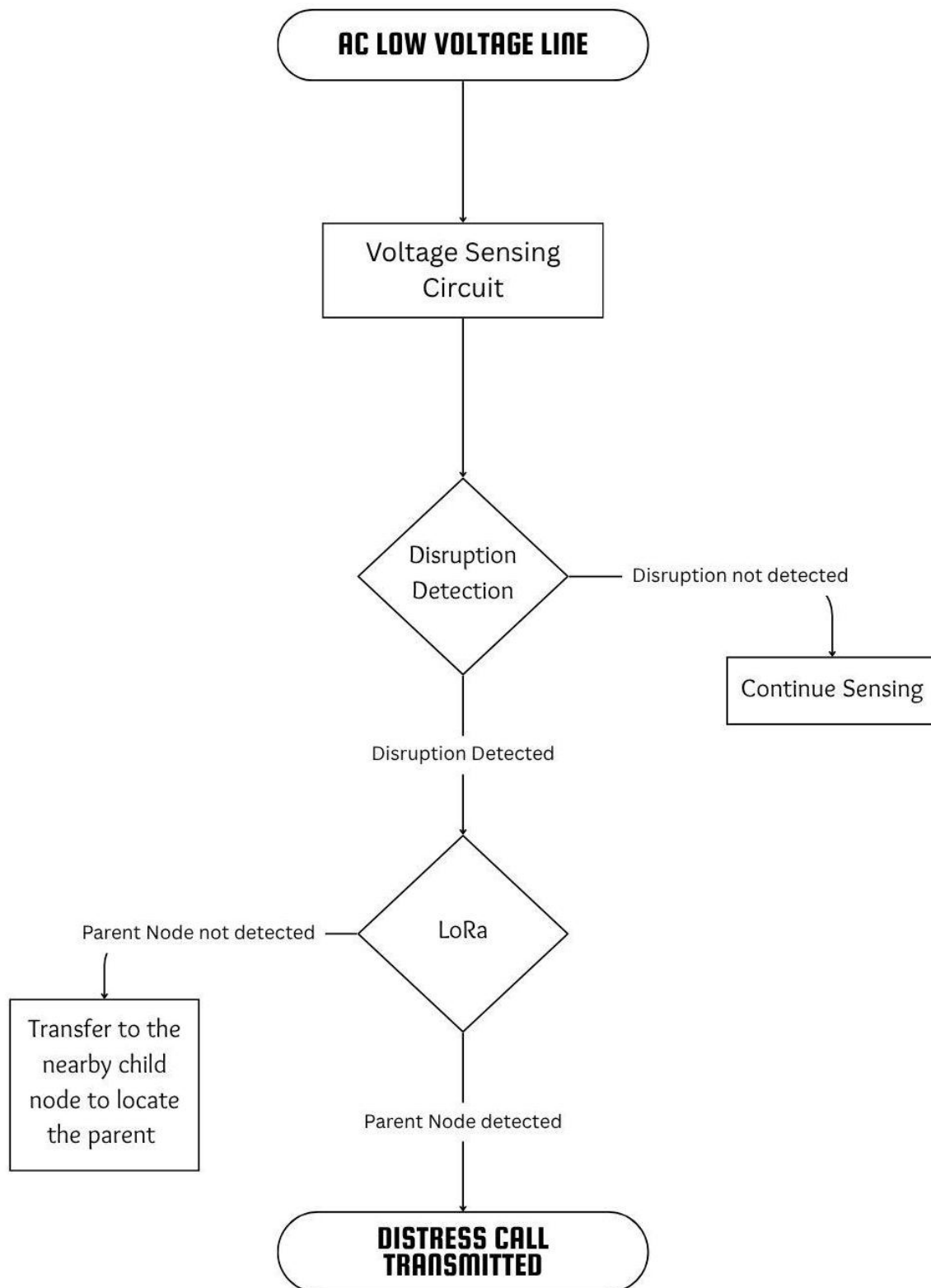


Fig-1: Child Module Functional Diagram & Flowchart

2. **Parent Node:** When a distress signal from any child node from its network is received, the parent node will send a signal to the solenoid actuator or Motorized MCCB to cut off the power line, with the help of the microcontroller. Ultimately shutting the whole system down, until the engineer appears to revive the fault.

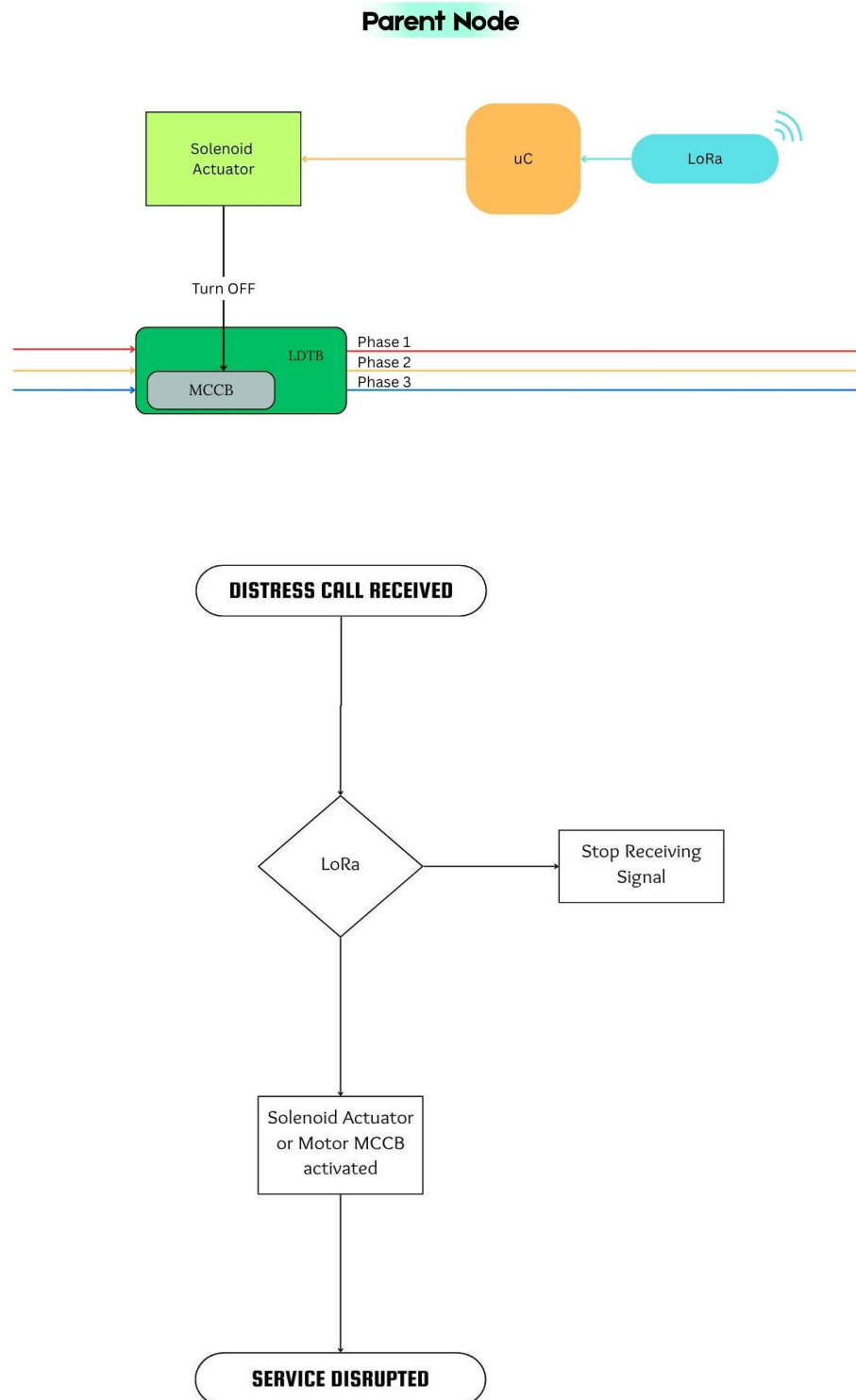


Fig-2: Parent Module Functional Diagram & Flowchart

Structural Representation:

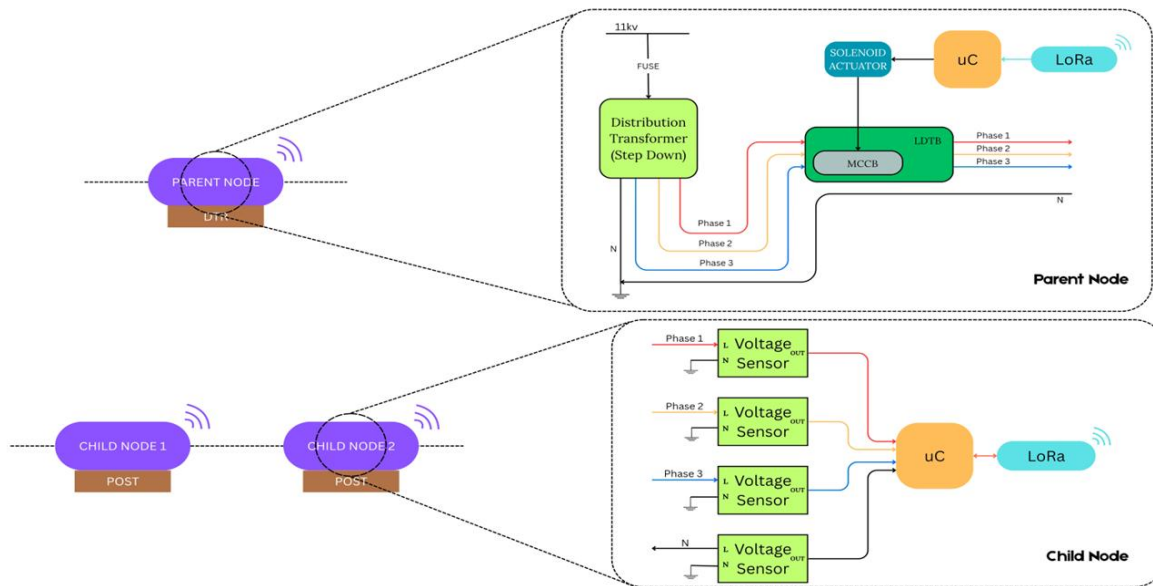


Fig-3: Equipments and Components of Parent and Child Module

Visual Representation:

Let's deep dive into the whole system. Consider the grid in fig-4a, which portrays the AC low voltage overhead wire grid system in which the poles are connected and further it is connected to the DTR of the sub station. Each pole consists of one child node module and the parent module is present in the substation DTR.

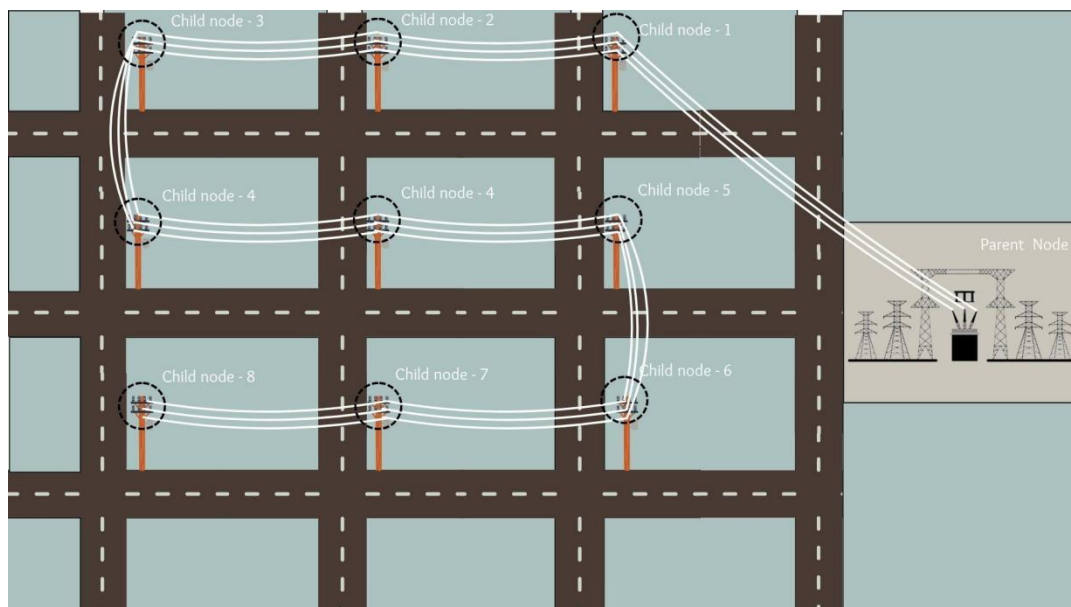


Fig-4a: How will the system work?(The network)

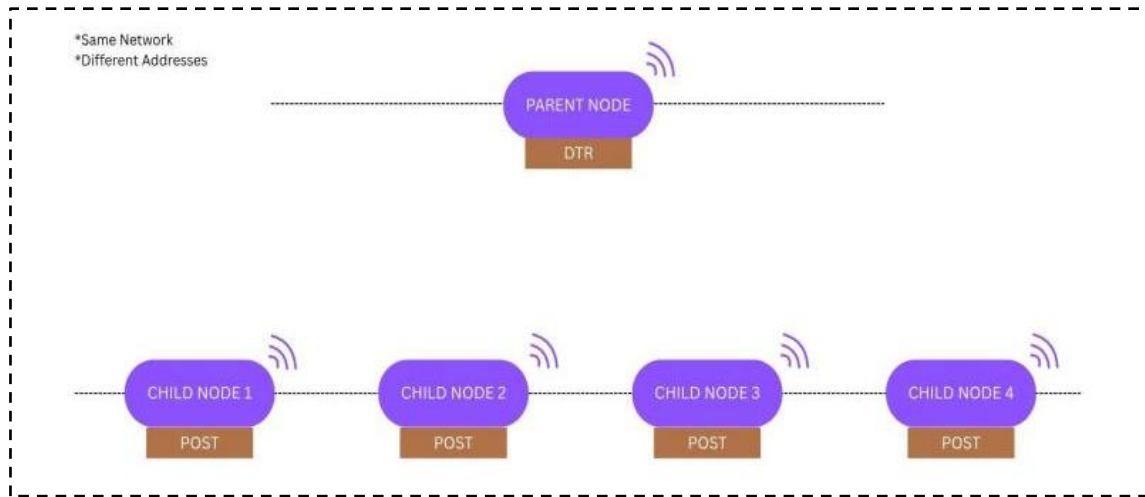


Fig-4b: Array's connectivity to the parent module

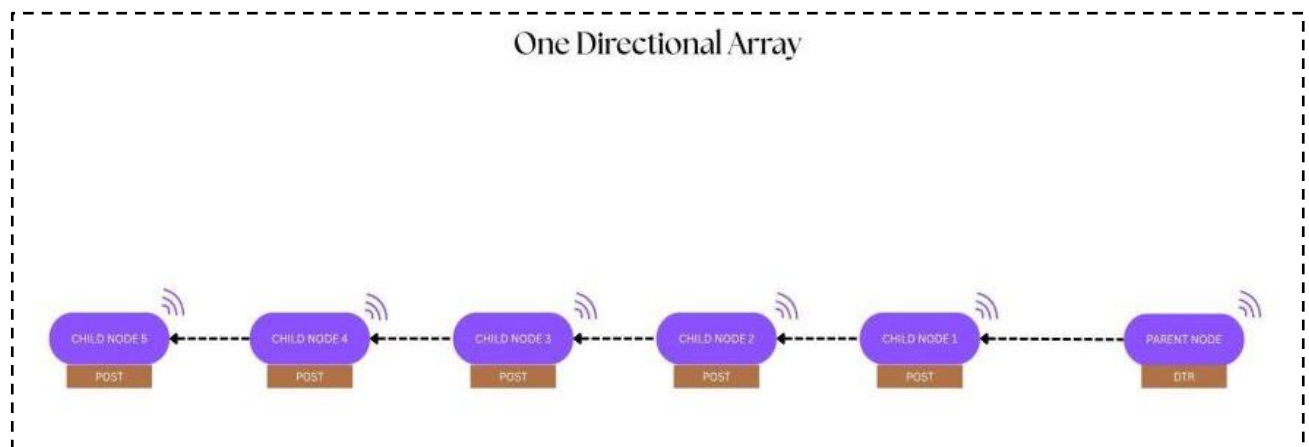


Fig-4c: Array's connectivity to the parent module

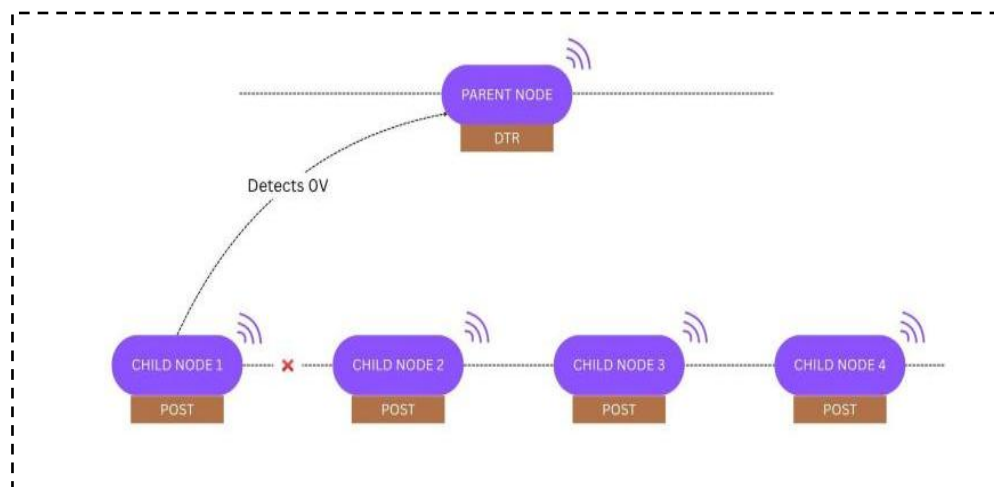


Fig-4d: Communication between the parent to a distressed child module

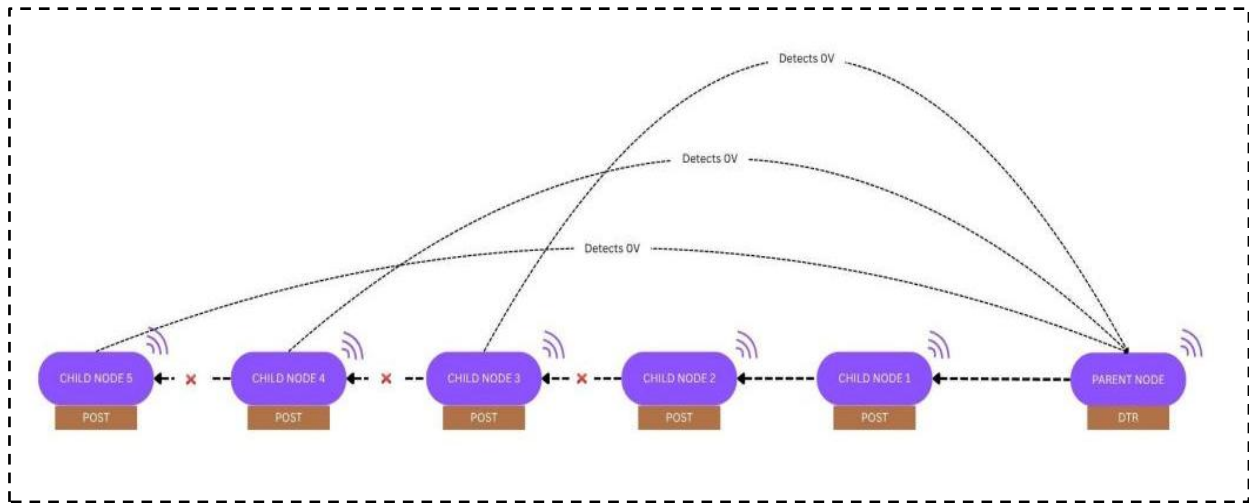


Fig-4e: Multiple Child nodes sending distress call

HOW THE SYSTEM WORKS:

Step 1: Continuous Monitoring: Each child node keeps checking the voltage of all three phases and the neutral line at regular intervals.

Step 2: Fault Identification: If the child node detects abnormal voltage patterns—such as a sudden drop, wire break, or imbalance—it marks it as a potential fault.

Step 3: Local Confirmation: To avoid false alarms, the child node rechecks the abnormal readings for a short confirmation period.

Step 4: Distress Message from Child Node: Once confirmed, the child node prepares a distress packet (fault type, node ID, timestamp, voltage values) and sends it wirelessly (LoRa) to the parent node.

Step 5: Fault Reception at Parent Node: The parent node receives the distress message and validates it to ensure it's genuine and not a communication error.

Step 6: Automatic Isolation: If the fault is confirmed, the parent node activates the relay system to trip the MCB and isolate the faulty section.

Step 7: Forwarding the Alert: Immediately after isolation, the parent node uploads the fault report to the cloud server.

Step 8: Notification to Electricity Board: The electricity board/maintenance team receives instant notifications through Dashboard alert, Message or Mobile Application.

Step 9: Confirmation to Field Team: The system logs the fault location and details so technicians can quickly identify and reach the exact affected point.

LORA NETWORK DISTRIBUTION:

Before diving deep into LoRa Network Distribution, first we have to know how LoRa modules communicate with each other. LoRa modules need to be connected in a specific network. A developer can set a LoRa module to a specific network represented by a specific numerical digit. Then it can connect to the LoRa modules which are connected to that exact same network. This is how they communicate. A LoRa having a Network ID can't connect to another LoRa which is not set with that particular ID.

In our system LoRa Network distribution will be a bit tricky. Because if a child module finds a parent module which is not actually its parent but it has the same network ID, it will turn off a healthy grid meanwhile the disputed one will still have power. This is a faulty situation that can happen if not taken care of.

Well we do have a solution for this. Let's look at this image -

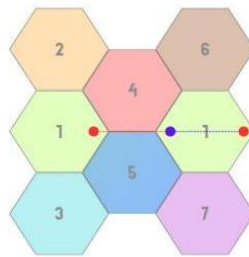


Fig-5: Representation of faulty condition

Here the parent node having the network ID - 1 is equidistant from both of the networks. So the child from both the network can connect to it making it a faulty network system. Let's look at below image -

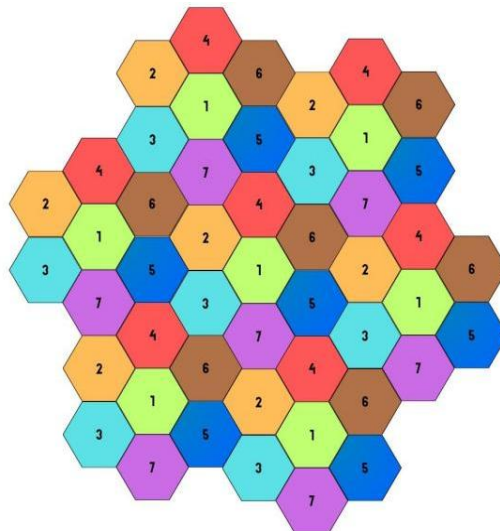


Fig-6: LoRa Module Distribution Network Design

In this image consider every hexagon as a LoRa network. In every hexagon LoRa network ID is the same. So as per our proposal the network will consist of one parent and multiple child nodes (The number of child nodes depends upon the scale of the distribution network). Now look at the numbers inside of every hexagon. This design is a proposal to avoid the aforementioned situation. Here every number denotes a unique network ID for each network. According to this design we are considering 7 network IDs (1-7) . 1 at the centre and others circling around. But how can this design avoid the faulty situation?

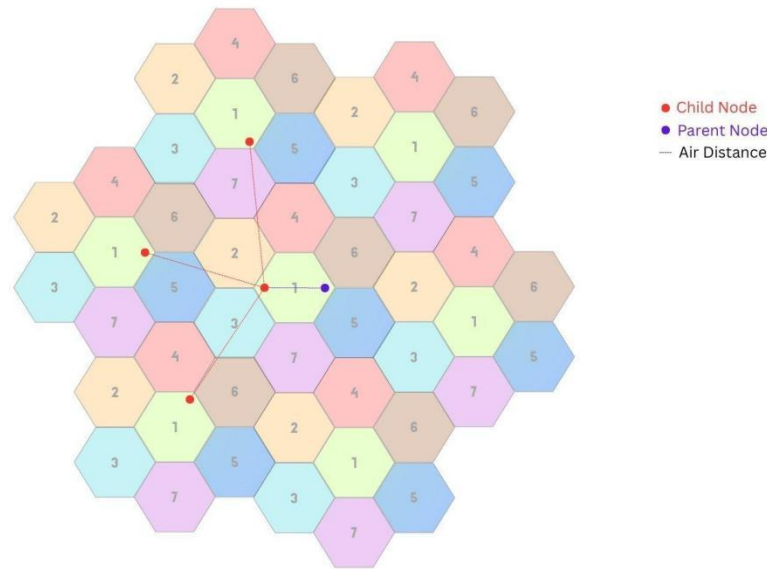


Fig-7: How the proposed design removes faulty situations

Now this image is showing four child nodes from Network 1 and one parent of network 1. But observe carefully the network at the middle of the image. If the child node isn't able to connect its parent node it must connect to the nearby child. But look, before reaching another network having network ID 1 it will get its own child nodes. So this design can avoid the faulty situation.

FEASIBILITY:

❖ Technical Feasibility:

- The project is equipped with technologies such as microcontrollers, AC voltage sensing modules, and low-power wireless communication modules (LoRa), which will be calibrated according to parameters.
- Use of mobile/cloud dashboards via platforms will ensure necessary data for the engineers

❖ Operational Feasibility:

- Modular design of child and parent nodes allows deployment in urban and rural settings, even where retrofitting is necessary, reducing modular cost
- The hardware complexity is low, needing less maintenance hardships.
- Wireless communication eliminates the need for costly rewiring, reducing the cost.

CRITICAL INFRASTRUCTURE HANDLING:

Sometimes some critical infrastructure like hospitals, fire stations often do share the same Distribution Transformer (DTR). This problem becomes critical when we shut off the entire area under that DTR. For that reason the system will have an Emergency node. This emergency node will also carry a LoRa module. Before tripping off the MCCB the parent node will notify the Emergency node by a Pre-Trip call to turn on the backup power generator. After notifying it will take a buffer time of 5sec and then finally cut-off the power. Thus this emergency handling system will continue providing power to emergency sector while keeping the affected area safe.

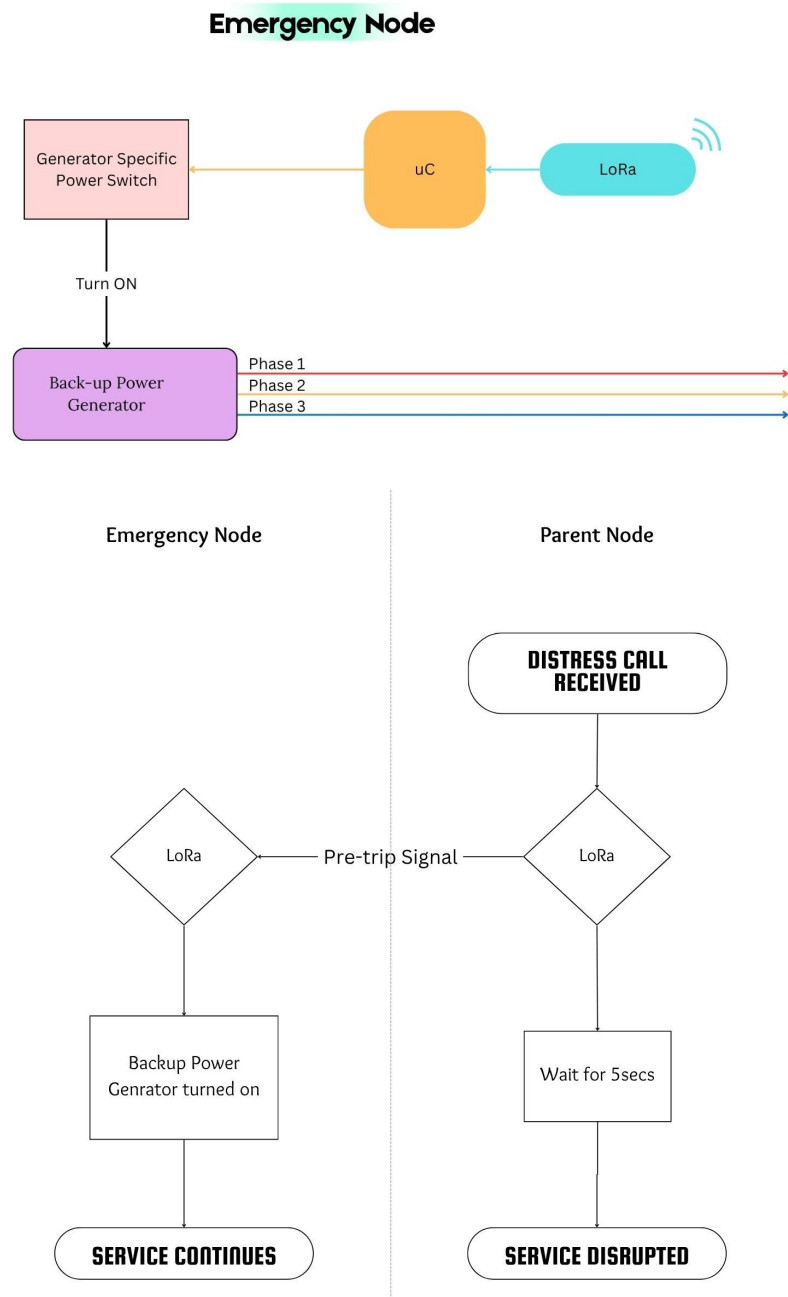


Fig-8: Emergency Module Functional Diagram & Flowchart

SCIENCE AND WORKING PRINCIPLE BEHIND THE IDEA

Electrical distribution faults, such as wire breakage or short circuits, typically lead to abrupt voltage drops, spikes or power interruptions in the AC low voltage lines(50V to 1KV), which affects the hosts (e.g.household properties, unaware pedestrians, street animals) of the power lines.

When an electrical wire breaks or there is a fault on a power line, the voltage in that line suddenly changes—it might drop sharply, spike unexpectedly, or even disappear altogether. Our system takes advantage of this by constantly monitoring the voltage in each of the three wires and the neutral line at different points in the power distribution network.

- **Continuous Voltage Monitoring:**

Voltage sensors measure the voltage on all three-phase lines and neutral conductors and calculate the potential difference between the nodes. They use well designed circuits that rely on electromagnetic induction and voltage transformer principles. This process converts high-voltage AC signals into low-voltage analog levels that can be measured.

- **Analog-to-Digital Conversion and Thresholding:**

The system converts the analog voltages to digital voltage with high-resolution ADCs in microcontrollers by using signal processing techniques to compare real-time voltage values against pre-defined safe thresholds(~ 230V). This helps detect any deviations that might indicate faults, using classic electrical engineering ideas about voltage stability and fault current behavior.

- **Wireless Fault Alert Transmission:**

When a fault is detected, the system forms a communication packet that represents the fault status and node identity. It sends this packet using low-power wide-area network protocols like LoRa, which uses spread-spectrum and chirp modulation techniques for strong, interference-resistant long-distance transmission(approximately 10-20 kms).

- **Fault Isolation via Motorized MCCB:**

The parent node receives fault data and checks the addressing and error-checking code to confirm the fault's validity. It then triggers the motorized MCCB. This driver energizes a coil, causing the MCCB to trip, which isolates the affected line segment quickly with latency ~120 ms to 700 ms

- **Real-Time Notification and Analytics:**

Data is sent to cloud-hosted dashboards through secure IP networking layers using REST APIs. This system allows power utility operators to monitor grid health, track fault incidents with graphical data presentation, and schedule field operations effectively.

- **Safety and Reliability:**

By automating the detection, isolation, and notification process using control systems, electrical signal theory, and wireless communications, the system improves response time, reduces human error, and boosts grid stability.

WHO WOULD BE BENEFICIARY OF THIS INNOVATION AND WHY?

- 1. Grid operators and electrical engineers:** Real time fault detection, message alerts in mobile and computers and low latency will help them to take immediate action on the faulty AC low voltage line and revive to normal conditions.
- 2. Residential and Industrial:** Gain from enhanced safety, uninterrupted power supply, and quick restoration after faults.
- 3. Rural and Remote Communities:** Wireless IOT enabled modules reduce infrastructure cost and enable coverage of hard to reach areas.
- 4. Common People:** Ensuring safety from electrocution during natural calamities (like flood) from human unidentified faulty AC LV wires.
- 5. Government and Regulatory Bodies:** Tracking and optimization of energy delivery, supporting environmental and economic policies.

FEATURE	EXISTING SOLUTION 'A' (WIRED SYSTEM)	EXISTING SOLUTION 'B' (GSM BASED SYSTEM)	DIVYA SYSTEM
Wireless Communication	✗	✓	✓
Real-Time Automated Detection	✗	✗	✓
Low Power Consumption	✗	✗	✓
Long Range(10+km)	✗	✓(dependent on coverage)	✓
Easy Scalability	✗	✗	✓
Cost Effective	✗	✗	✓
Sustainable for Rural Deployment	✗	Limited	✓

BUILDING A PROTOTYPE AND TESTING

Materials Chart with cost:

Serial No.	DETAILS			COST
	NAME	SPECIFICATION	QUANTITY	
1	LoRa		3	₹3,000.00
2	Display	OLED	1	₹100.00
3	Voltage Sensor	ZMPT101B	4	₹200.00
4	Solenoid Actuator	JF-0630B - 24V	1	₹300.00

The entire system will cost roughly around Rs.5000 which is absolute cheapest in it's genre.

For prototype we have used nRF24LO1 module which provides wireless connection just like LoRa but in a less range. We have used Arduino Uno as a microcontroller to test the system. We have achieved a milestone - the prototype works properly just what we expected.

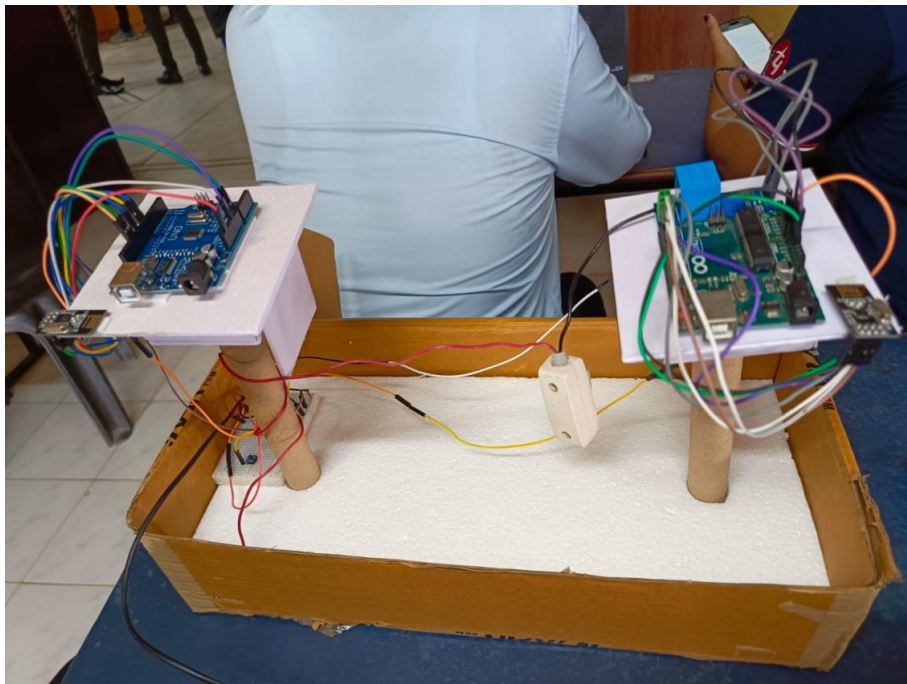


Fig-9: Prototype of 'Divya System'

FINAL OUTCOME OF THE PROJECT

The final outcome of the project is a solution for all the faulty AC Low Voltage overhead wires, which will reduce the death risk due to electrocution and will ensure speedy restoration of the faulty lines.

- **Integrated Fault Detection and Isolation System:**

A fully functional, automated solution comprising child nodes equipped with voltage sensors and wireless transmitters, and a parent node that receives fault alerts and controls fault isolation devices practically ready for deployment on electrical distribution networks.

- **Real-Time Fault Monitoring:**

Continuous measurement and analysis of three-phase and neutral voltages, enabling instantaneous detection of wire breakage or hazardous voltage abnormalities.

- **Wireless Communication Setup:**

Reliable low-power, long-range wireless transmission (using LoRa) between distributed sensor nodes and the central parent node ensures flexible, scalable installation without the need for additional wiring infrastructure.

- **Automated Fault Isolation:**

The parent node activates the motorized MCCB which will trip instantly upon fault detection, minimizing safety hazards like electrocution and preventing spread of faults.

- **Mobile and Cloud Monitoring Dashboard:**

Real-time dashboards with historical logs and push notifications provide utility operators and maintenance teams with actionable insights for rapid response and strategic planning.

- **Prototype Validation and Field-Ready Design:**

The system will be practically validated with the microcontroller (Arduino Nano), voltage sensors, and communication modules (LoRa). It will include detailed wiring, firmware, and software components designed for main production.

- **Latency and Safety Compliance:**

The solution ensures fault detection-to-isolation latency within fewer than 1 second, suitable for safe electrical distribution practices as per industry standards.

- **Cost-Efficient and Modular Architecture:**

Employs commercially available, low-cost components with the new and innovative parent module-child module and child module-child module architecture which will be effective towards the fault detection.